

CO3-AT3-Debugging and Optimization

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QUESTIONS

Q1ATM Withdrawal – Exception
Handling Debugging

10 marks

**Q2**Hospital Registration –
NullPointerException
Debugging

10 marks

**Q3**E-Commerce Product Search –
Array Exception Debugging

10 marks

**Q4**Bank Account – Race Condition
Debugging

10 marks

**Q5**Railway Reservation –
Synchronization Debugging

Q2 Hospital Registration – NullPointerException Debugging

10 marks

A hospital patient registration application crashes when patient information is unavailable.

Debug the following complete Java program.

Task:

1. Identify the runtime error.
2. Modify the program to handle a missing patient name.
3. Handle invalid age input.
4. Use try-catch-finally appropriately.
5. The program must always display "Registration completed".
6. Submit the COMPLETE corrected Java program.

Faulty Program:

```
import java.util.Scanner;

class PatientRegistration {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);
```

10 marks

5/5 answered

```
System.out.print("Enter patient name: ");
String inputName = sc.nextLine();

String name;

if (inputName.equals("NULL")) {
    name = null;
} else {
    name = inputName;
}

System.out.print("Enter age: ");
int age = sc.nextInt();

System.out.println("Patient Name: "
    + name.toUpperCase());

if (age >= 18) {
    System.out.println("Adult patient");
} else {
    System.out.println("Minor patient");
}

System.out.println("Registration completed");
}
}
```

The corrected program must prevent NullPointerException and handle invalid age input without terminating unexpectedly.

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```

1 import java.util.Scanner;
2 import java.util.InputMismatchException;
3
4 class PatientRegistration{
5     public static void main (String[] args){
6         Scanner sc = new Scanner(System.in);
7
8         try {
9             System.out.print("Enter patient name :")
10             String inputName = sc.nextLine();
11
12             String name;
13
14             if(inputName.equals("NULL")){
15                 name = null;
16
17             }else{
18                 name = inputName;
19             }
20             System.out.println("Enter age :");
21             int age = sc.nextInt();
22
23             if(name == null){
24                 System.out.println("Patient Name : ")
25             }else{
26                 System.out.println("Patient Name :")
27             }
28

```

Test Input

HARI
19

Output

Enter patient name
:Enter age :
Patient Name :HARI
Adult patient
registered completed :

🕒 Pending manual review

```
29         if(age>=18) {
30             System.out.println("Adult patient")
31         }else{
32             System.out.println("Minor patient")
33         }
34     }catch(InputMismatchException e) {
35         System.out.println("Invalid age input")
36     }
37     catch(Exception e) {
38         System.out.println("Unexpected erro
39     }finally{
40         System.out.println("registered comp
41         sc.close();
42     }
43 }
44
45 }
```

 Submitted

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